



DEPARTMENT OF COMPUTING
AND CONTROL ENGINEERING

ДЕПАРТМАН ЗА РАЧУНАРСТВО
И АУТОМАТИКУ

ACSE

Odsek za automatiku, geomatiku
i upravljanje sistemima

Sub- department for Automatic Control
and System Engineering

Vremenski diskretni modeli

Modeliranje i simulacija sistema

Upravljanje, modelovanje i simulacija sistema

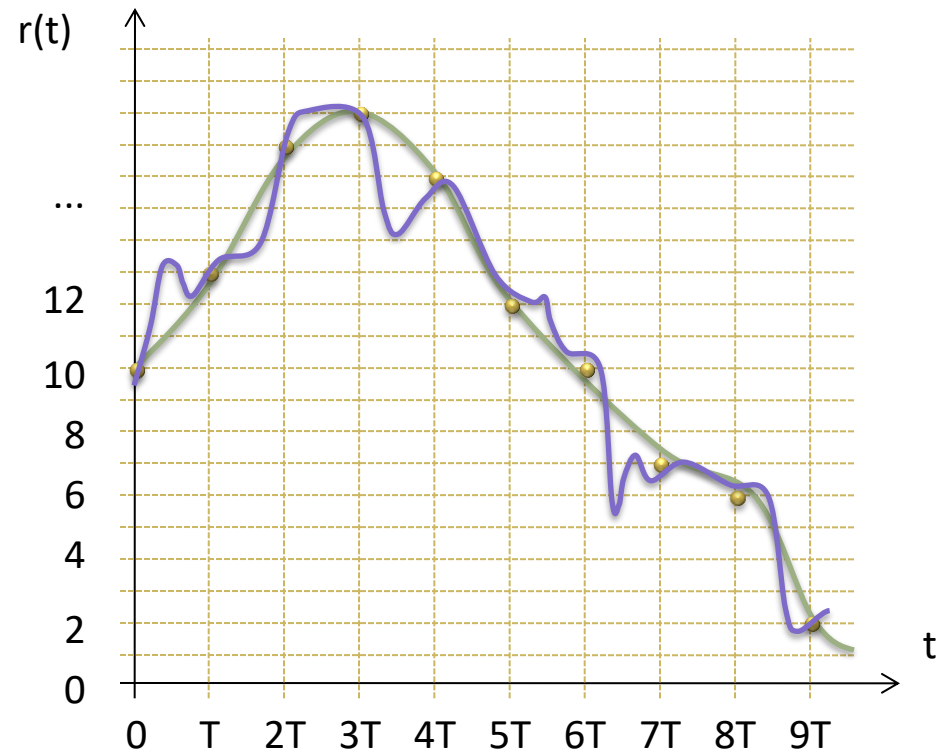
Digitalan model

- Razlozi upotrebe digitalnih modela
 - Upotreba računara (obrada)
 - Kodovanje signala
 - Prenos signala na daljinu bez smetnji
 - Vremenski multipleks
- Ovde posmatramo formiranje digitalnog (vremenski diskretnog) linearnog modela u prostoru stanja na osnovu vremenski kontinualnog linearnog modela u prostoru stanja

Kvantovanje signala

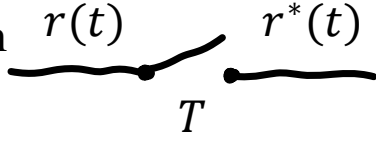
- Signali se predstavljaju brojnim vrednostima
- Signali se diskretizuju
 - Po vremenu
 - Po nivou (kvantovanje po nivou)
- Kvantovanje po vremenu vrši „odabirač“
 - Na izlazu se pojavljuje povorka impulsa (odbiraka) u trenucima odabiranja $t=kT$, $k=0,1,2,\dots$
 - Perioda odabiranja T
- Kvantovanje po nivou vrši A/D konvertor (analogno-digitalni)
 - Na izlazu se dobijaju brojne vrednosti
 - Broj nivoa (kvantova) zavisi od rezolucije A/D konvertora
 - $2^8, 2^{10}, 2^{12}, 2^{16}$
 - Mi ćemo pretpostaviti da je rezolucija A/D konvertora dovoljno velika

Kvantovanje signala - primer

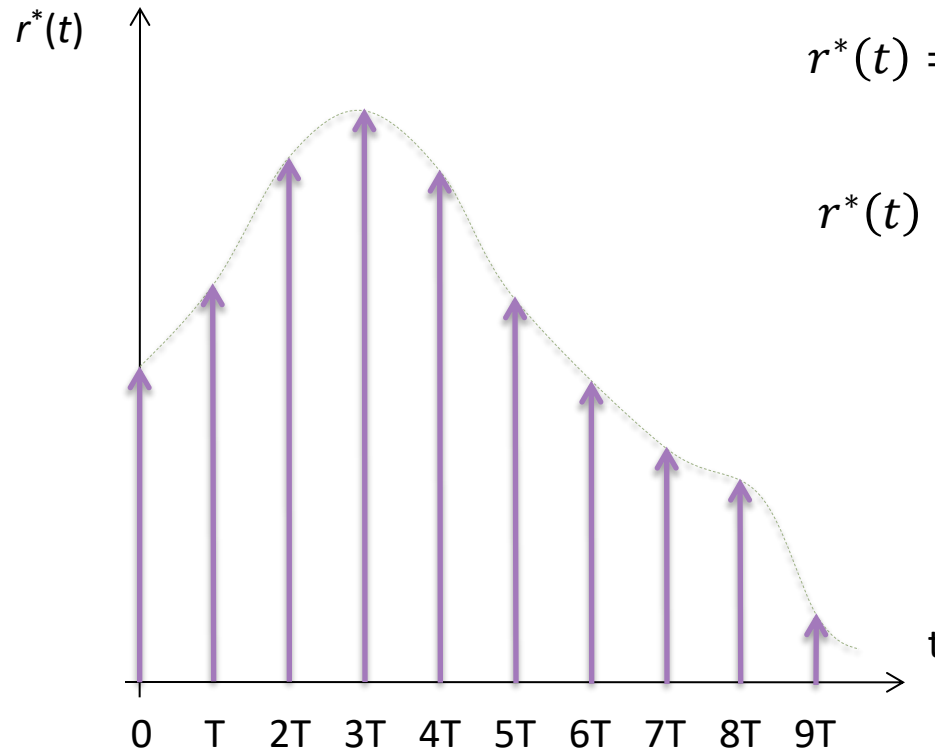


$r^* = [10 \ 13 \ 17 \ 18 \ 16 \ 12 \ 10 \ 7 \ 6 \ 2]$

Odabirač

Kontinualan signal $r(t)$  Vremenski diskretan signal $r^*(t)$

$$r^*(t) = \begin{cases} r(kT), & t = kT \\ 0, & t \neq kT \end{cases} \quad k = 0, 1, \dots$$

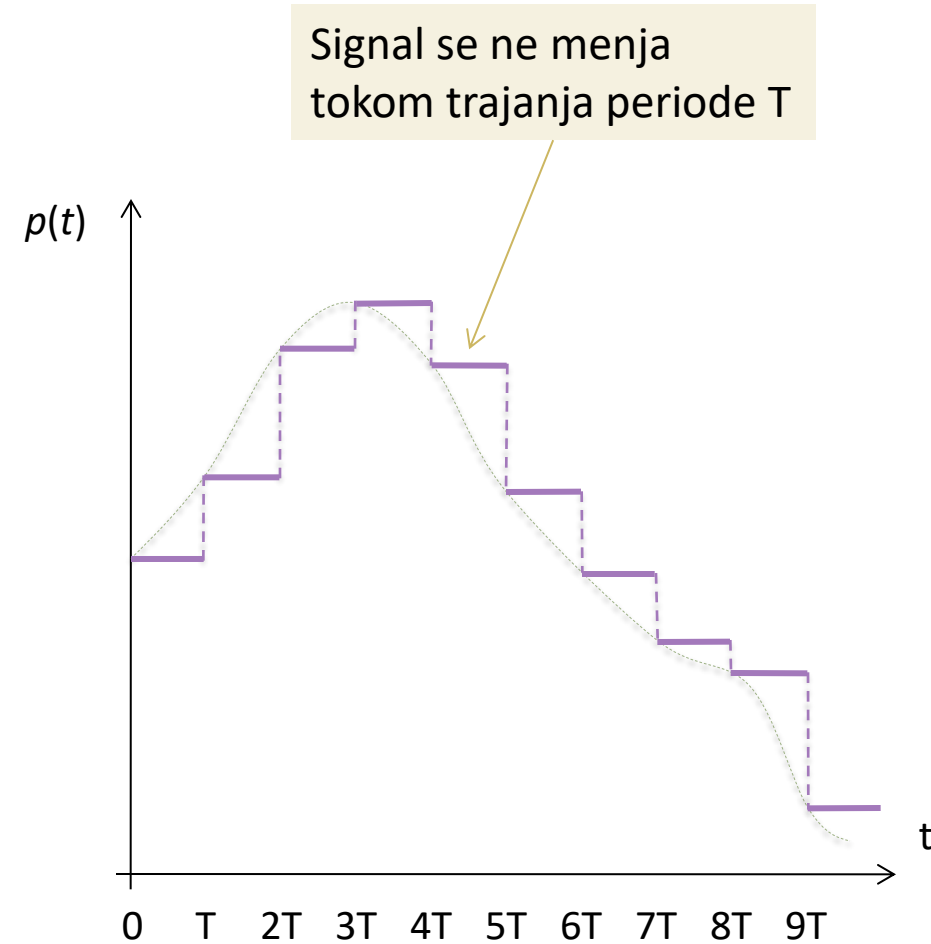
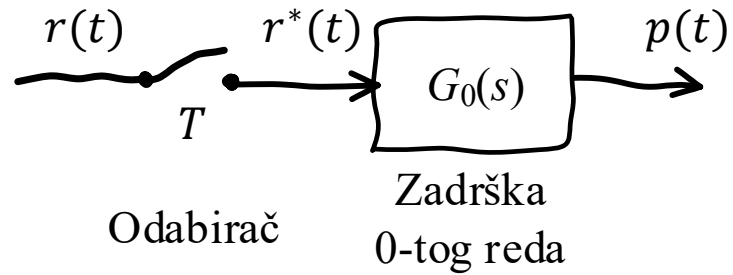


$$r^*(t) = r(kT) \cdot \delta(t - kT), \quad k = 0, 1, \dots$$

$$r^*(t) = \sum_{k=0}^{\infty} r(kT) \cdot \delta(t - kT)$$

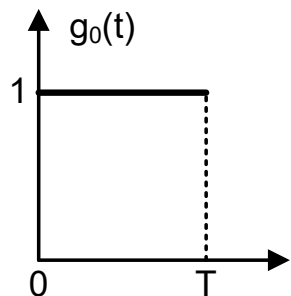
Kolo zadržke nultog reda

- Realizacija procesa odabiranja vrži i zadržku signala



Model zadržke nultog reda

Jedan odbirak:



$$g_0(t) = h(t) - h(t - T)$$

$$G_0(s) = \frac{1}{s} - \frac{e^{-sT}}{s} = \frac{1 - e^{-sT}}{s}$$

Model zadržke

Ceo diskretizovan signal:

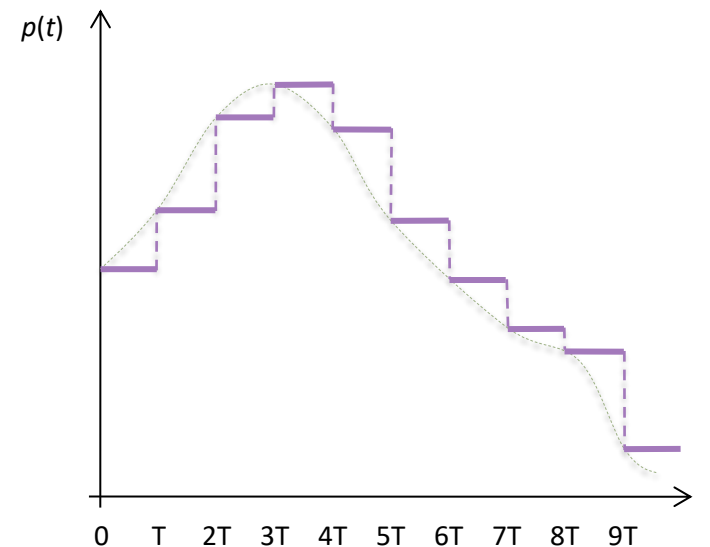
$$p(t) = \sum_{k=0}^{\infty} r(kT) \cdot \{h(t - kT) - h(t - (k+1)T)\} \quad / L$$

$$P(s) = \sum_{k=0}^{\infty} r(kT) \cdot \left\{ \frac{e^{-kTs}}{s} - \frac{e^{-(k+1)Ts}}{s} \right\}$$

$$P(s) = \sum_{k=0}^{\infty} r(kT) \cdot \frac{1 - e^{-sT}}{s} \cdot e^{-kTs} = \frac{1 - e^{-sT}}{s} \cdot \sum_{k=0}^{\infty} r(kT) \cdot e^{-kTs}$$

$$P(s) = G_0(s) \cdot R^*(s)$$

$$R^*(s) = \sum_{k=0}^{\infty} r(kT) \cdot e^{-kTs}$$



Kompleksan lik
povorke odbiraka $r^*(t)$

Podsećanje:
čisto vremensko kašnjenje:
 $\mathcal{L}\{f(t - \tau)\} = e^{-s\tau}F(s)$

Z-transformacija

- Povorka vremenski diskretizovanog signala i njegova Laplasova transformacija

$$r^*(t) = \sum_{k=0}^{\infty} r(kT) \cdot \delta(t - kT)$$

$$R^*(s) = \mathcal{L}\{r^*(t)\} = \sum_{k=0}^{\infty} r(kT) \cdot \mathcal{L}\{\delta(t - kT)\} = \sum_{k=0}^{\infty} r(kT) \cdot e^{-kTs}$$

- Smena uvodi novu kompleksnu promenljivu z
 $z = e^{sT}$

- Z-transformacija povorke signala

$$\mathcal{Z}\{r(t)\} = \mathcal{Z}\{r^*(t)\} = R(z)$$

$$R(z) = \sum_{k=0}^{\infty} r(kT) \cdot z^{-k}$$

Čisto vremensko kašnjenje:

$$\mathcal{L}\{f(t - T)\} = e^{-sT} F(s)$$

za

$$z = e^{sT}$$

dobija se

$$\mathcal{Z}\{f(t - T)\} = z^{-1} F(z)$$

Za $t = kT, k = 0, 1, 2, \dots$

$$\mathcal{Z}\{f(kT)\} = F(z)$$

Ili kraće

$$\mathcal{Z}\{f(k)\} = F(z)$$

pa je

$$\mathcal{Z}\{f(k - 1)\} = z^{-1} F(z)$$

$$\mathcal{Z}\{f(k - 2)\} = z^{-2} F(z)$$

...

$$\mathcal{Z}\{f(k + 2)\} = z^2 F(z)$$

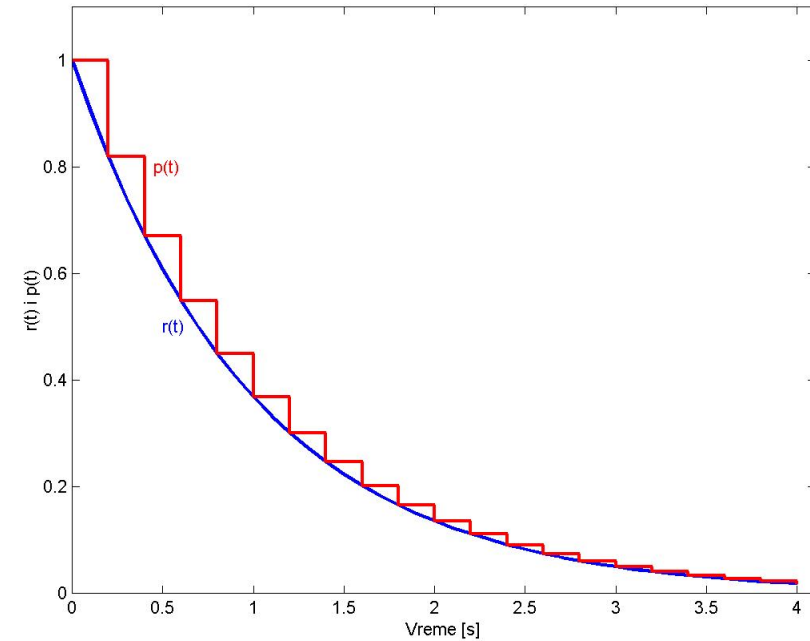
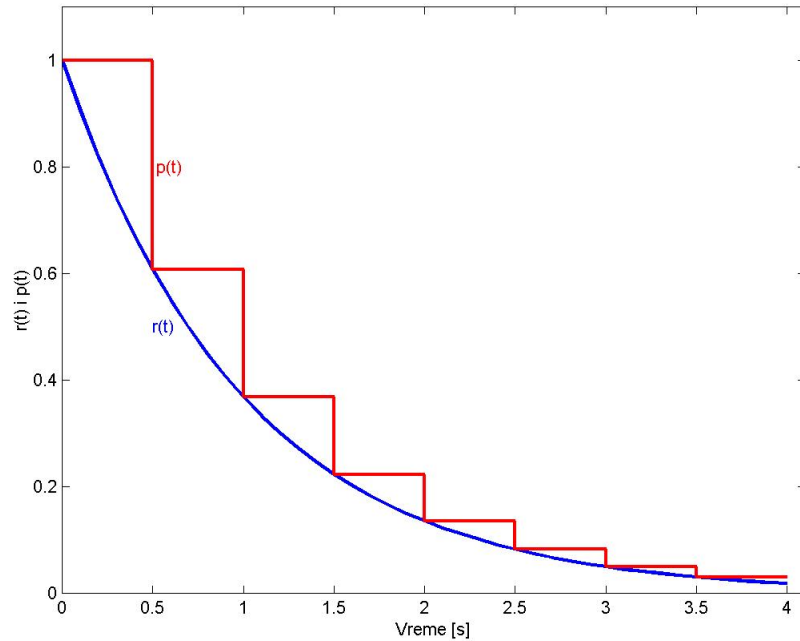
...

Teorema o odabiranju

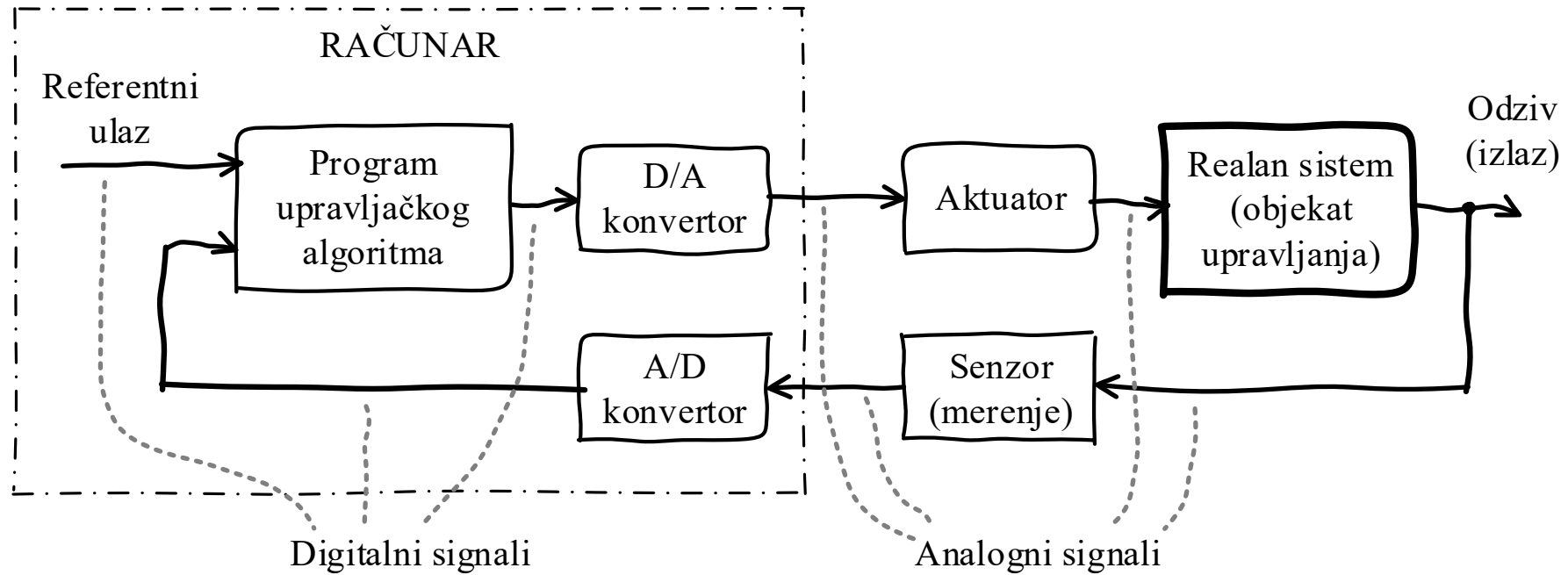
- Ako kontinualan signal $f(t)$ ne sadrži harmonike u području učestanosti $> \omega_0$ [rad/s], on se može kompletno okarakterisati vrednostima signala merenim u trenucima međusobno udaljenim za vreme $T = 0.5(\frac{2\pi}{\omega_0})$.
- Teorijski: Perioda uzorkovanje signala treba da je barem 2 puta kraća od periode (prostoperiodične) komponente signala koja ima najveću učestanost
- Praktično: Perioda uzorkovanja treba da je 10-20 puta kraća

Primer odabiranja

- Primer signala odabiranog različitim periodama odabiranja
 - $T_a=0.5s$
 - $T_b=0.2s$



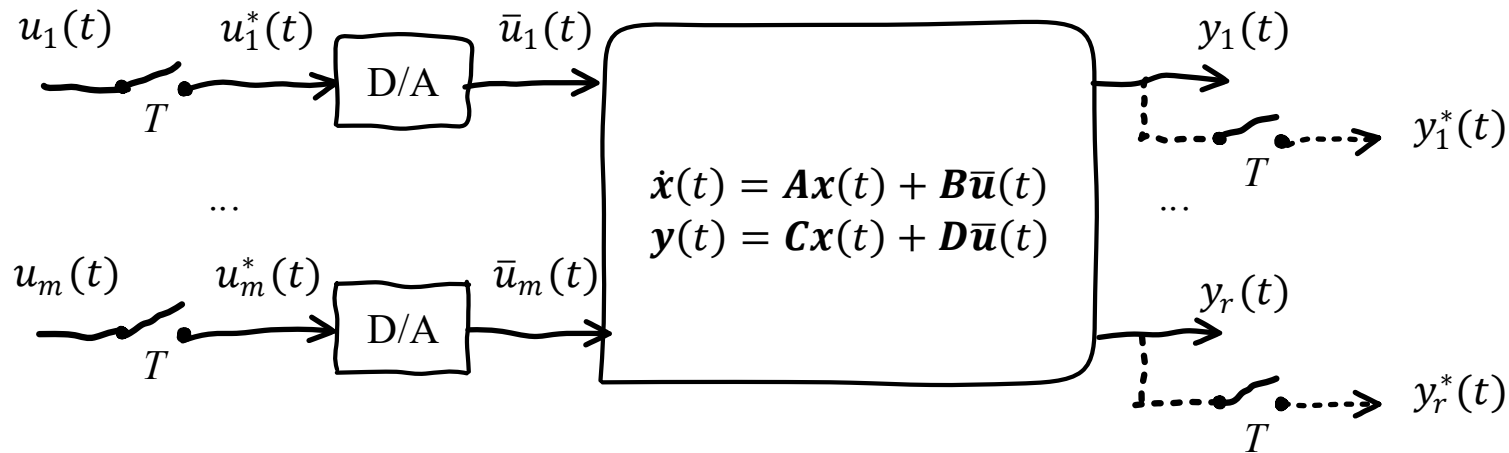
Upravljanje računarom



Odziv na diskretnu pobudu

Posmatra se vremenski kontinualan model ponašanja objekta upravljanja, ali ...

1. Tokom jedne periode T ulazi $u(t)$ su konstantni
2. Izlazi sistema nas interesuju samo u trenucima odabiranja $t = kT$



$$\mathbf{x}(k+1) = \mathbf{E} \cdot \mathbf{x}(k) + \mathbf{F} \cdot \mathbf{u}(k), \quad \mathbf{x}(0) = \mathbf{x}_0$$

$$\mathbf{y}(k) = \mathbf{C} \cdot \mathbf{x}(k) + \mathbf{D} \cdot \mathbf{u}(k)$$

$$k \equiv kT, \quad k = 0, 1, 2, \dots$$

Linearni modeli u prostoru stanja

Izvod po vremenu

$$\dot{\mathbf{x}}(t) = \mathbf{A} \cdot \mathbf{x}(t) + \mathbf{B} \cdot \mathbf{u}(t), \quad \mathbf{x}(0) = \mathbf{x}_0$$
$$\mathbf{y}(t) = \mathbf{C} \cdot \mathbf{x}(t) + \mathbf{D} \cdot \mathbf{u}(t)$$

← Linearan **kontinualan**
matematički model
u prostoru stanja

Nema izvoda, nego se određuje vrednost u narednom trenutku

$$\mathbf{x}(k + 1) = \mathbf{E} \cdot \mathbf{x}(k) + \mathbf{F} \cdot \mathbf{u}(k), \quad \mathbf{x}(0) = \mathbf{x}_0$$
$$\mathbf{y}(k) = \mathbf{C} \cdot \mathbf{x}(k) + \mathbf{D} \cdot \mathbf{u}(k)$$
$$k \equiv kT, \quad k = 0, 1, 2, \dots$$

← Linearan **vremenski
diskretan**
matematički model
u prostoru stanja

Računanje diskretanog odziva

- Vrednosti promenljivih stanja u diskretnim trenucima se mogu izračunati rekurzivno na osnovu poznavanja $\mathbf{x}(0)$ i vrednosti ulaza u svim trenucima $\mathbf{u}(k)$, $k = 0, 1, 2, \dots$

$$\mathbf{x}(1) = \mathbf{E} \cdot \mathbf{x}(0) + \mathbf{F} \cdot \mathbf{u}(0)$$

$$\mathbf{x}(2) = \mathbf{E} \cdot \mathbf{x}(1) + \mathbf{F} \cdot \mathbf{u}(1)$$

$$\mathbf{x}(3) = \mathbf{E} \cdot \mathbf{x}(2) + \mathbf{F} \cdot \mathbf{u}(2)$$

...

$$\mathbf{x}(k + 1) = \mathbf{E} \cdot \mathbf{x}(k) + \mathbf{F} \cdot \mathbf{u}(k)$$

Vreme,
tj. diskretni vremenski trenuci
 $t = kT, \quad k = 0, 1, 2, \dots$



Kontinualan odziv

- Posmatramo linearan matematički model u prostoru stanja

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A} \cdot \mathbf{x}(t) + \mathbf{B} \cdot \mathbf{u}(t), & \mathbf{x}(0) &= \mathbf{x}_0 & (*) \\ \mathbf{y}(t) &= \mathbf{C} \cdot \mathbf{x}(t) + \mathbf{D} \cdot \mathbf{u}(t)\end{aligned}$$

← Linearan kontinualan matematički model u prostoru stanja

- Odziv modela je

$$\begin{aligned}\mathbf{x}(t) &= \Phi(t) \cdot \mathbf{x}(0) + \int_0^t \Phi(t - \tau) \cdot \mathbf{B} \cdot \mathbf{u}(\tau) d\tau, & 0 \leq t & & (**) \\ \mathbf{y}(t) &= \mathbf{C} \cdot \mathbf{x}(t) + \mathbf{D} \cdot \mathbf{u}(t)\end{aligned}$$

gde je $\Phi(t)$ fundamentalna matrica sistema $\Phi(t) = e^{At}$

- Izvođenje:

- Nakon primene Laplasove transformacije na (*) dobija se

$$\begin{aligned}s\mathbf{X}(s) - \mathbf{x}(0) &= \mathbf{A} \cdot \mathbf{X}(s) + \mathbf{B} \cdot \mathbf{U}(s) \\ s\mathbf{X}(s) - \mathbf{A} \cdot \mathbf{X}(s) &= (s\mathbf{I} - \mathbf{A}) \cdot \mathbf{X}(s) = \mathbf{x}(0) + \mathbf{B} \cdot \mathbf{U}(s) \\ \mathbf{X}(s) &= (s\mathbf{I} - \mathbf{A})^{-1} \cdot \mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1} \cdot \mathbf{B} \cdot \mathbf{U}(s)\end{aligned}$$

- Primenom inverzne Laplasove transformacije na $\mathbf{X}(s)$ dobija se (**)

$$\Phi(t) = L^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$$

Primer računanja fundamentalne matrice

(ovaj primer je ranije rađen, a sada samo podsećanje)

$$A = \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad C = [0 \quad 5] \quad D = 0$$

$$R(s) = (sI - A)^{-1} = \begin{bmatrix} s+3 & 2 \\ -1 & s \end{bmatrix}^{-1} = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s & -2 \\ 1 & s+3 \end{bmatrix}$$

$$R(s) = \begin{bmatrix} \frac{-1}{s+1} + \frac{2}{s+2} & \frac{-2}{s+1} + \frac{2}{s+2} \\ \frac{1}{s+1} + \frac{-1}{s+2} & \frac{2}{s+1} + \frac{-1}{s+2} \end{bmatrix}$$

$$\Phi(t) = L^{-1}\{R(s)\} = \begin{bmatrix} -e^{-t} + 2e^{-2t} & -2e^{-t} + 2e^{-2t} \\ e^{-t} - e^{-2t} & 2e^{-t} - e^{-2t} \end{bmatrix}$$

Vremenski diskretan odziv

- Na osnovu (**) odziv u trenutku $t = kT$ je

$$(**) \quad \mathbf{x}(t) = \mathbf{\Phi}(t) \cdot \mathbf{x}(0) + \int_0^t \mathbf{\Phi}(t - \tau) \cdot \mathbf{B} \cdot \mathbf{u}(\tau) d\tau$$

$$\mathbf{x}(kT) = \mathbf{\Phi}(kT) \cdot \mathbf{x}(0) + \int_0^{kT} \mathbf{\Phi}(kT - \tau) \cdot \mathbf{B} \cdot \mathbf{u}(\tau) d\tau$$

- a u narednom trenutku $t = kT + T$ je

$$\mathbf{x}(kT + T) = \mathbf{\Phi}(kT + T) \cdot \mathbf{x}(0) + \int_0^{kT} \mathbf{\Phi}(kT + T - \tau) \cdot \mathbf{B} \cdot \mathbf{u}(\tau) d\tau + \int_{kT}^{kT+T} \mathbf{\Phi}(kT + T - \tau) \cdot \mathbf{B} \cdot \mathbf{u}(\tau) d\tau$$

- Na osnovu osobine: $\mathbf{\Phi}(a + b) = \mathbf{\Phi}(a) \cdot \mathbf{\Phi}(b)$

$$\mathbf{x}(kT + T) = \mathbf{\Phi}(T) \cdot \left(\mathbf{\Phi}(kT) \cdot \mathbf{x}(0) + \int_0^{kT} \mathbf{\Phi}(kT - \tau) \cdot \mathbf{B} \cdot \mathbf{u}(\tau) d\tau \right) + \int_{kT}^{kT+T} \mathbf{\Phi}(kT + T - \tau) \cdot \mathbf{B} \cdot \mathbf{u}(\tau) d\tau$$

- period $[kT, kT + T]$ je dovoljno kratak da možemo smatrati da je $\mathbf{u}(kT)$ nepromenljivo tokom tog perioda.

$$\mathbf{x}(kT + T) = \mathbf{\Phi}(T) \cdot \mathbf{x}(kT) + \left(\int_{kT}^{kT+T} \mathbf{\Phi}(kT + T - \tau) d\tau \right) \cdot \mathbf{B} \cdot \mathbf{u}(kT)$$

...

Vremenski diskretan odziv (2)

- Rešavanje se nastavlja uvođenjem smena ...

- Prva smena: $q = \tau - kT$

$$\int_{kT}^{kT+T} \Phi(kT + T - \tau) d\tau = \int_0^T \Phi(T - q) dq$$

- i još jedna smena: $y = T - q$

$$\int_0^T \Phi(T - q) dq = - \int_T^0 \Phi(y) dy = \int_0^T \Phi(y) dy$$

- Konačno:

$$\mathbf{x}(kT + T) = \Phi(T) \cdot \mathbf{x}(kT) + \left(\int_0^T \Phi(t) dt \right) \cdot \mathbf{B} \cdot \mathbf{u}(kT)$$

tj.

$$\mathbf{x}(k + 1) = \mathbf{E} \cdot \mathbf{x}(k) + \mathbf{F} \cdot \mathbf{u}(k)$$

$$\mathbf{E} = \Phi(T)$$

$$\mathbf{F} = \left(\int_0^T \Phi(t) dt \right) \cdot \mathbf{B}$$

Primer: Linearan vremenski kontinualan matematički model u prostoru stanja

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t), \quad \mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$y(t) = [0 \quad 5] \mathbf{x}(t)$$

Matrice modela su

$$\mathbf{A} = \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{C} = [0 \quad 5], \quad \mathbf{D} = 0$$

Fundamentalna matrica

$$\Phi(t) = \begin{bmatrix} -e^{-t} + 2e^{-2t} & -2e^{-t} + 2e^{-2t} \\ e^{-t} - e^{-2t} & 2e^{-t} - e^{-2t} \end{bmatrix}$$

Vremenska diskretizacija

$$\mathbf{E}(T) = \Phi(T) = \begin{bmatrix} -e^{-T} + 2e^{-2T} & -2e^{-T} + 2e^{-2T} \\ e^{-T} - e^{-2T} & 2e^{-T} - e^{-2T} \end{bmatrix}$$
$$\mathbf{F}(T) = \left(\int_0^T \Phi(t) dt \right) \mathbf{B} = \begin{bmatrix} \int_0^T (-e^{-t} + 2e^{-2t}) dt \\ \int_0^T (e^{-t} - e^{-2t}) dt \end{bmatrix} = \begin{bmatrix} e^{-T} + e^{-2T} \\ \frac{1}{2} - e^{-T} + \frac{1}{2} e^{-2T} \end{bmatrix}$$

Za $T = 0.1$ dobija se:

$$\mathbf{E} = \begin{bmatrix} 0.7326 & -0.1722 \\ 0.0861 & 0.9909 \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} 0.0861 \\ 0.0045 \end{bmatrix}$$

Linearan vremenski diskretan model u prostoru stanja

$$\mathbf{x}(k+1) = \begin{bmatrix} 0.7326 & -0.1722 \\ 0.0861 & 0.9909 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0.0861 \\ 0.0045 \end{bmatrix} u(k)$$
$$y(k) = [0 \quad 5] \mathbf{x}(k)$$

Kako da izračunamo matrice E i F?

- Tražimo pogodnu transformaciju $\mathbf{x}(t) = \mathbf{P} \cdot \tilde{\mathbf{x}}(t)$
tako da model možemo opisati kao $\dot{\tilde{\mathbf{x}}}(t) = \tilde{\mathbf{A}} \cdot \tilde{\mathbf{x}}(t) + \tilde{\mathbf{B}} \cdot \mathbf{u}(t)$

- Ako je $\tilde{\mathbf{A}} = \begin{bmatrix} \lambda_1 & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \lambda_n \end{bmatrix}$, tada je $\tilde{\Phi}(t) = \begin{bmatrix} e^{\lambda_1 t} & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & e^{\lambda_n t} \end{bmatrix}$

- Veza sa $\mathbf{A}$:

$$\dot{\mathbf{x}}(t) = \mathbf{A} \cdot \mathbf{x}(t) + \mathbf{B} \cdot \mathbf{u}(t)$$

$$\dot{\tilde{\mathbf{x}}}(t) = \mathbf{P}^{-1} \cdot \dot{\mathbf{x}}(t) = \mathbf{P}^{-1} \cdot (\mathbf{A} \cdot \mathbf{x}(t) + \mathbf{B} \cdot \mathbf{u}(t)) = \mathbf{P}^{-1} \cdot \mathbf{A} \cdot \mathbf{x}(t) + \mathbf{P}^{-1} \cdot \mathbf{B} \cdot \mathbf{u}(t)$$

$$= \underbrace{\mathbf{P}^{-1} \cdot \mathbf{A} \cdot \mathbf{P}}_{\tilde{\mathbf{A}}} \cdot \tilde{\mathbf{x}}(t) + \underbrace{\mathbf{P}^{-1} \cdot \mathbf{B}}_{\tilde{\mathbf{B}}} \cdot \mathbf{u}(t)$$

$\tilde{\mathbf{A}}$

$\tilde{\mathbf{B}}$

- daje

$$\mathbf{A} \cdot \mathbf{P} = \mathbf{P} \cdot \tilde{\mathbf{A}}$$

Svojstvene vrednosti matrice

- Jednačina: $\mathbf{A} \cdot \mathbf{P} = \mathbf{P} \cdot \tilde{\mathbf{A}}$
se može napisati kao:

$$\mathbf{A} \cdot [\mathbf{p}_1 \quad \mathbf{p}_2 \quad \dots \quad \mathbf{p}_n] = [\mathbf{p}_1 \quad \mathbf{p}_2 \quad \dots \quad \mathbf{p}_n] \cdot \begin{bmatrix} \lambda_1 & & & \mathbf{0} \\ & \lambda_2 & & \\ & & \ddots & \\ \mathbf{0} & & & \lambda_n \end{bmatrix}$$

λ_i su svojstvene vrednosti matrice $\mathbf{A}$

$\mathbf{p}_i$ su svojstveni vektori matrice $\mathbf{A}$

- Tj. može se predstaviti kao sistem jednačina

$$\mathbf{A} \cdot \mathbf{p}_i = \mathbf{p}_i \lambda_i, \quad i = 1, 2, \dots, n$$

ili

$$(\mathbf{A} - \lambda_i \mathbf{I}) \cdot \mathbf{p}_i = \mathbf{0}, \quad i = 1, 2, \dots, n$$

gde je netrivialno rešenje za:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

Primer

$$\mathbf{A} = \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix}$$

$$\det(\lambda \mathbf{I} - \mathbf{A}) = (\lambda + 3)\lambda + 2 = \lambda^2 + 3\lambda + 2 = 0 \rightarrow \lambda_1 = -1, \lambda_2 = -2 : \quad \tilde{\mathbf{A}} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$$

$$\mathbf{A}\mathbf{p}_1 = \lambda_1 \mathbf{p}_1$$

$$\begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} p_{11} \\ p_{12} \end{bmatrix} = - \begin{bmatrix} p_{11} \\ p_{12} \end{bmatrix} \rightarrow p_{11} = -p_{12}, \quad \text{npr. } p_{11} = 1, p_{12} = -1$$

$$\mathbf{A}\mathbf{p}_2 = \lambda_2 \mathbf{p}_2$$

$$\begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} p_{21} \\ p_{22} \end{bmatrix} = -2 \begin{bmatrix} p_{21} \\ p_{22} \end{bmatrix} \rightarrow p_{21} = -2p_{22}, \quad \text{npr. } p_{21} = -2, p_{22} = 1$$

$$\mathbf{P} = [\mathbf{p}_1 \quad \mathbf{p}_2] = \begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix}, \quad \mathbf{P}^{-1} = \begin{bmatrix} -1 & -2 \\ -1 & -1 \end{bmatrix}$$

$$\text{Provera: } \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{bmatrix} -1 & -2 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} = \tilde{\mathbf{A}}$$

Računanje E i F

- Kada nađemo svojstvene vrednosti i svojstvene vektore matrice A , tada možemo izračunati E i F :

$$\tilde{x}(k+1) = \tilde{\Phi}(T)\tilde{x}(k) + \dots$$

$$\begin{aligned} x(k+1) &= P\tilde{x}(k+1) = P\tilde{\Phi}(T)\tilde{x}(k) + \dots = \\ &= P\tilde{\Phi}(T)P^{-1} \cdot x(k) + \dots = \Phi(T)x(k) + \dots \end{aligned}$$

$$E = \Phi(T) = P\tilde{\Phi}(T)P^{-1}$$

$$\begin{aligned} F &= \left(\int_0^T \Phi(t) dt \right) B = \left(\int_0^T P\tilde{\Phi}(t)P^{-1} dt \right) B = P \left(\int_0^T \tilde{\Phi}(t) dt \right) P^{-1} B \\ &= P\tilde{A}^{-1}(\tilde{\Phi}(T) - I)P^{-1} B \end{aligned}$$

jer je:

$$\int_0^T \tilde{\Phi}(t) dt = \begin{bmatrix} \int_0^T e^{\lambda_1 t} dt & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \int_0^T e^{\lambda_n t} dt \end{bmatrix} = A^{-1}(\Phi(T) - I)$$

$$\frac{1}{\lambda_1} e^{\lambda_1 t} \Big|_0^T = \frac{1}{\lambda_1} (e^{\lambda_1 T} - 1)$$

$$\begin{aligned} x(k+1) &= E \cdot x(k) + F \cdot u(k) \\ E &= \Phi(T) \\ F &= \left(\int_0^T \Phi(t) dt \right) \cdot B \end{aligned}$$

$$x(t) = P \cdot \tilde{x}(t)$$

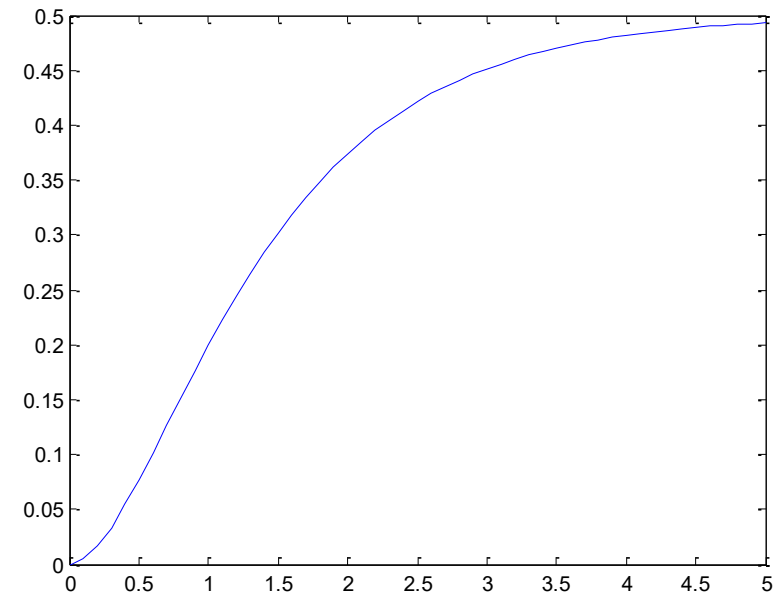
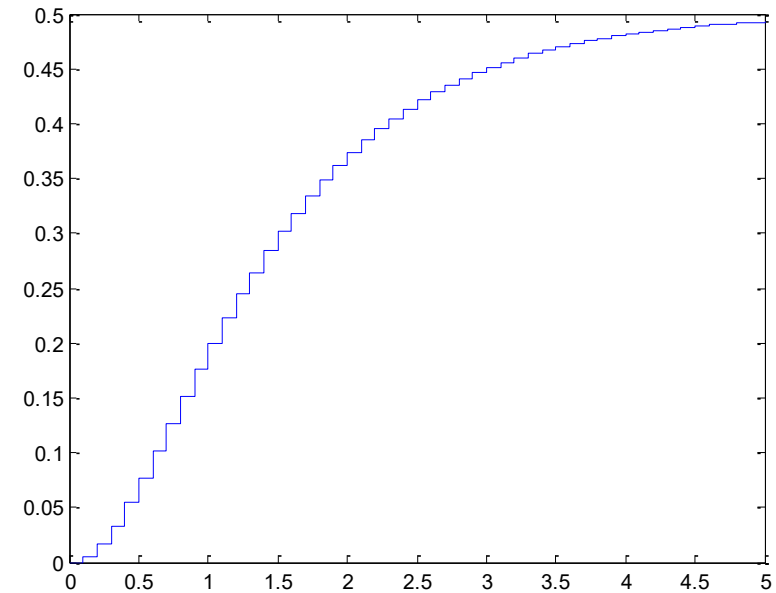
Primer 1 (Matlab)

```
>> A = [-3 -2; 1 0];
>> B = [1; 0];
>> C = [0 1];
>> D = 0;

>> [P,lambda]=eig(A)
P =
    -0.8944    0.7071
     0.4472   -0.7071
lambda =
     -2      0
      0     -1

>> Ts = 0.1;           % perioda odabiranja
>> E = P*expm(lambda*Ts)/P
E =
     0.7326    -0.1722
     0.0861     0.9909
>> F = P/lambda*(expm(lambda*Ts)-eye(2))/P * B
F =
     0.0861
     0.0045

>> u = ones(1,5/Ts+1);
>> x(:,1) = [0;0];    % =x0
>> for k=1:length(u)-1
        x(:,k+1) = E*x(:,k) + F*u(k);
    end
>> y = C*x + D*u;
>> t = (0:length(y)-1)*Ts;
>> stairs(t,y)         % crtanje sa zadržkom
>> plot(t,y)           % ... kao da je kontinualan
```



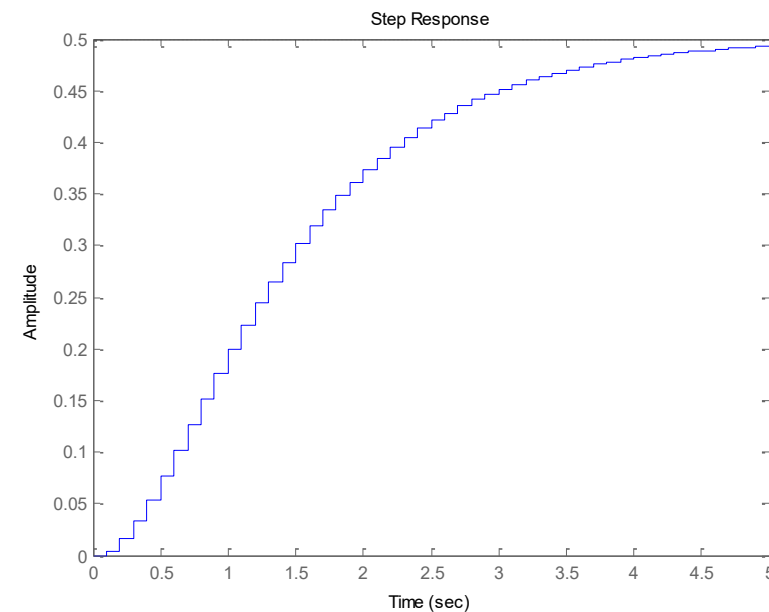
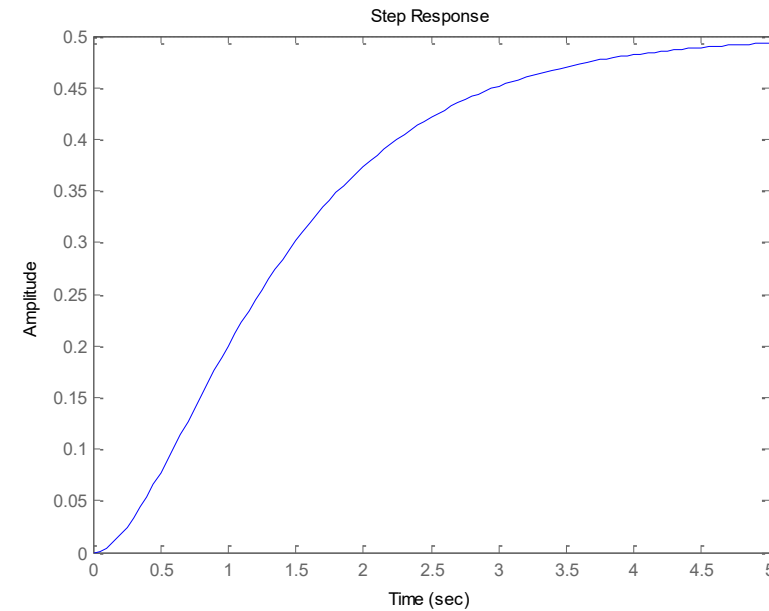
Primer 1 (2)

```
>> A = P*lambda/P % provera
A =
    -3    -2
     1     0

>> m=ss(A,B,C,D); % kont.model
>> md=c2d(m,Ts) % diskretizacija
a =
           x1      x2
    x1    0.7326   -0.1722
    x2    0.08611   0.9909
b =
           u1
    x1    0.08611
    x2    0.004528
c =
           x1  x2
    y1     0   1
d =
           u1
    y1     0
Sampling time: 0.1
Discrete-time model.

>> expm(A*Ts) % E=fundamentalna matrica
ans =
    0.7326   -0.1722
    0.0861   0.9909

>> step(m,5) % odziv kont.modela
>> step(md,5) % odziv diskretnog modela
```



Primer 1 (Julija)

```
using LinearAlgebra
A = [-3.0 -2.0; 1.0  0.0]
B = [1.0; 0.0]
C = [0.0  1.0]
D = 0.0

λ = eigvals(A)      # sopstvene vrednosti
P = eigvecs(A)      # sopstveni vektori

# provera
@show P * Diagonal(λ) / P

Ts = 0.1             # perioda odabiranja
E = P * Diagonal(exp.(λ*Ts)) / P
F = P * Diagonal((exp.(λ*Ts) .- 1.0) ./ λ) / P * B
```

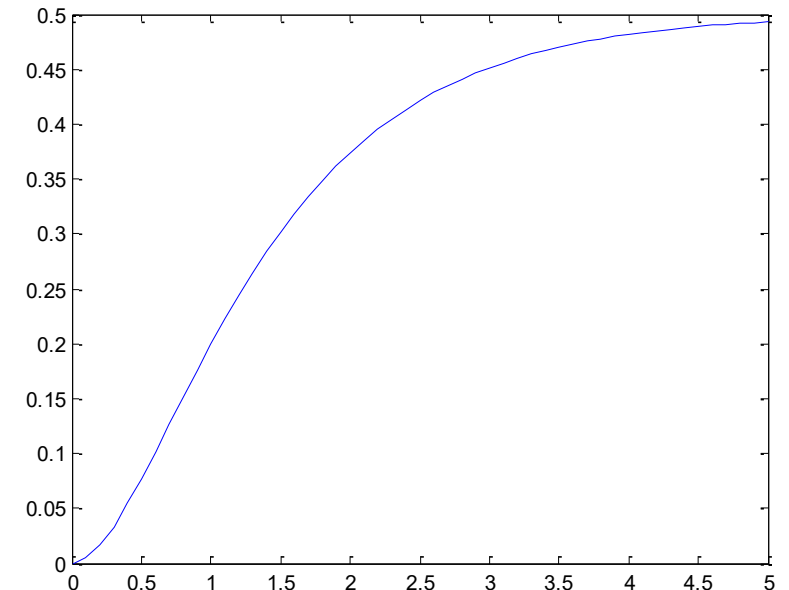
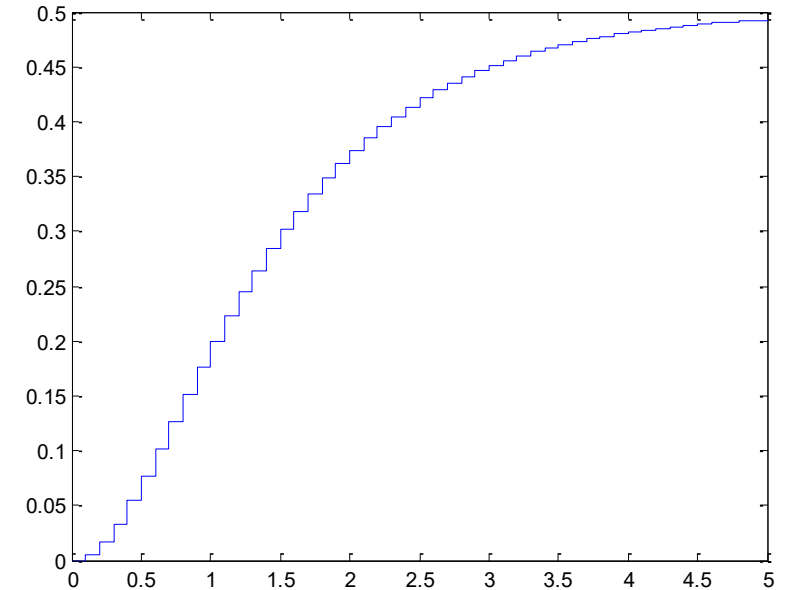
```
# simulacija
n = Int(5/Ts+1)
u = ones(n)
x0 = [0.0; 0.0]
x = x0
X = zeros(n,2)
X[1,:] = x'           # x0
for k = 1:(n-1)
    global x = E*x + F*u[k]
    global X[k+1,:] = x'
end
Y = C*X' + D*u'
t = (0:n-1)*Ts
plot(t,Y',linetype=:steppre)
plot(t,Y')
```

```
# simulacija paketom ControlSystems
using ControlSystems

m = ss(A,B,C,D)        # kont.model
md,_ = c2d(m,Ts)        # diskretizacija

y,t,x = step(m,5)       # odziv kont.modela
yd,td,x = step(md,5)    # odziv diskretnog modela
plot(t,y)
plot(td,yd,linetype=:steppre)
```

```
# crtanje sa zadržkom
# ... kao da je kontinualan
```



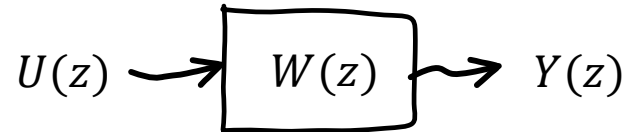
Funkcija diskretnog prenosa

- Funkcija diskretnog prenosa je odnos Laplasove transformacije (kompleksnih likova) **vremenski diskretizovanih** signala izlaza i ulaza, za nulte početne vrednosti

$$\frac{Y(z)}{U(z)} = W(z)$$

← funkcija diskretnog prenosa
 $W(z)$

- Simbolički prikaz



- Funkcija diskretnog prenosa se realizuje kao količnik polinoma po z

$$W(z) = \frac{b_m z^m + b_{m-1} z^{m-1} + \dots + b_2 z^2 + b_1 z + b_0}{z^n + a_{n-1} z^{n-1} + \dots + a_2 z^2 + a_1 z + a_0}$$

ili

$$W(z^{-1}) = z^{-(n-m)} \frac{b_m + b_{m-1} z^{-1} + \dots + b_1 z^{-(m-1)} + b_0 z^{-m}}{1 + a_{n-1} z^{-1} + \dots + a_1 z^{-(n-1)} + a_0 z^{-n}}$$

Diferencna jednačina

$$Y(z) = W(z)U(z) = \frac{B(z)}{A(z)}U(z) \Rightarrow A(z)Y(z) = B(z)U(z)$$

- U razvijenom obliku (uz: $d = n - m$)

$$\begin{aligned} & (1 + a_1z^{-1} + \dots + a_{n-1}z^{-(n-1)} + a_nz^{-n})Y(z) \\ & = z^{-d}(b_0 + b_1z^{-1} + \dots + b_{m-1}z^{-(m-1)} + b_mz^{-m})U(z) \end{aligned}$$

$$Y(z) + a_1z^{-1}Y(z) + \dots + a_nz^{-n}Y(z) = b_0z^{-d}U(z) + \dots + b_mz^{-d-m}U(z)$$

- Na osnovu osobine “pomeranje u vremenskom domenu”

$$y(k) + a_1y(k-1) + \dots + a_ny(k-n) = b_0u(k-d) + \dots + b_mu(k-d-m)$$

$$y(k) = -\sum_{i=0}^n a_i y(k-i) + \sum_{i=0}^m b_i u(k-d-i), \quad k \equiv kT, k = 0, 1, \dots$$

Diferencna
jednačina

Funkcija diskretnog prenosa

na osnovu vremenski diskretnog modela u prostoru stanja

- Primenimo Laplasovu transformaciju na vremenski diskretan model u prostoru stanja sa jednim ulazom i jednim izlazom

$$\begin{aligned}\mathbf{x}(k+1) &= \mathbf{E} \cdot \mathbf{x}(k) + \mathbf{F} \cdot u(k) \quad / \mathcal{L} \\ \mathbf{X}(s)e^{sT} &= \mathbf{E} \cdot \mathbf{X}(s) + \mathbf{F} \cdot U(s)\end{aligned}$$

- Ranije uvedena smena: $z = e^{sT}$

$$\begin{aligned}z \cdot \mathbf{X}(z) &= \mathbf{E} \cdot \mathbf{X}(z) + \mathbf{F} \cdot U(z) \\ (z\mathbf{I} - \mathbf{E}) \cdot \mathbf{X}(z) &= \mathbf{F} \cdot U(z) \\ \mathbf{X}(z) &= (z\mathbf{I} - \mathbf{E})^{-1} \cdot \mathbf{F} \cdot U(z)\end{aligned}$$

- Izlaz...

$$\begin{aligned}Y(z) &= \mathbf{C} \cdot \mathbf{X}(z) + \mathbf{D} \cdot U(z) \\ Y(z) &= \mathbf{C} \cdot (z\mathbf{I} - \mathbf{E})^{-1} \cdot \mathbf{F} \cdot U(z) + \mathbf{D} \cdot U(z)\end{aligned}$$

$$\frac{Y(z)}{U(z)} = W(z) = \mathbf{C} \cdot (z\mathbf{I} - \mathbf{E})^{-1} \cdot \mathbf{F} + \mathbf{D}$$

Primer

Matrice vremenski diskretnog modela su:

$$\mathbf{E} = \begin{bmatrix} 0.7326 & -0.1722 \\ 0.0861 & 0.9909 \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} 0.0861 \\ 0.0045 \end{bmatrix}, \quad \mathbf{C} = [0 \quad 5], \quad \mathbf{D} = 0$$

Funkcija diskretnog prenosa

$$\frac{Y(z)}{U(z)} = W(z) = \mathbf{C}(z\mathbf{I} - \mathbf{E})^{-1}\mathbf{F} + \mathbf{D}$$

$$W(z) = [0 \quad 5] \left(z \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.7326 & -0.1722 \\ 0.0861 & 0.9909 \end{bmatrix} \right)^{-1} \begin{bmatrix} 0.0861 \\ 0.0045 \end{bmatrix} + 0 = [0 \quad 5] \begin{bmatrix} z - 0.7326 & 0.1722 \\ -0.0861 & z - 0.9909 \end{bmatrix}^{-1} \begin{bmatrix} 0.0861 \\ 0.0045 \end{bmatrix}$$
$$W(z) = \frac{0.02264z + 0.02049}{z^2 - 1.724z + 0.7408} = \frac{0.02264z^{-1} + 0.02049z^{-2}}{1 - 1.724z^{-1} + 0.7408z^{-2}}$$

Diferencna jednačina

$$\frac{Y(z)}{U(z)} = \frac{0.02264z^{-1} + 0.02049z^{-2}}{1 - 1.724z^{-1} + 0.7408z^{-2}}$$

$$(1 - 1.724z^{-1} + 0.7408z^{-2})Y(z) = (0.02264z^{-1} + 0.02049z^{-2})U(z)$$

$$y(k) - 1.724y(k-1) + 0.7408y(k-2) = 0.02264u(k-1) + 0.02049u(k-2)$$